Nithish Reddy Bhoomannagari

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Summary

Recent MS Computer Science graduate (Oregon State, March 2026) who left industry to study interactive computer graphics with the goal of building technology for healthcare. I write C#, Python and SQL, work on Linux and in Docker, and write the tests that keep software reliable. At Oregon State's XR Lab I built and tested a Unity research tool and ran an IRB-approved 14-person user study. At Tata Consultancy Services I built Python and SQL pipelines, REST APIs and CI/CD release gates for 500+ stakeholders.

Education

Master of Science in Computer Science, Oregon State University, GPA: 3.7 *Sept 2023 – Mar 2026*
B.Tech, Electrical and Electronics Engineering, B. V. Raju Institute of Technology, CGPA: 8.06/10 *2016 – 2020*

Experience

Software Engineer | NexaNet, LLC Oct 2026 – Present
Python, FastAPI, PostgreSQL, scikit-learn, PyTorch, Docker, GitHub Actions *Cleveland, OH(Remote)*

- Built a FastAPI service in Python that serves website recommendations from a scikit-learn model, with typed Pydantic schemas.
- Moved the data layer from SQLite to PostgreSQL with SQLAlchemy, so the API could serve concurrent requests without locking.
- Added an evaluation endpoint reporting precision, recall and a confusion matrix on held-out data before each model release.
- Used GPT-4o with structured prompts to explain each recommendation in one line, validating outputs against a JSON schema.
- Containerized the service with Docker and documented the endpoints through FastAPI's OpenAPI docs for the team.

Software Engineer (Research Assistant) – EECS XR Lab | Oregon State University Dec 2023 – Mar 2026
C#, Unity 3D, Unity Test Framework, Unity Profiler, Docker, Linux, WSL2, GitHub *Corvallis, OR*

- Built a C# Unity tool that loads JSON exports of HPC networks as 3D graphs in VR and desktop, with sphere and force layouts.
- Wrote **66** functional test cases and automated the scene load, UI and interaction flows as Unity Play Mode tests.
- Fixed test crashes on simulated UI clicks by separating EventSystem and Input System events in the test harness.
- Fixed OpenGL failures in Docker on WSL2 by mapping the GPU and display sockets, so the Linux build ran on Windows hosts.
- Traced startup freezes with Unity Profiler and spread loading across frames, holding **60+ FPS** at **10K+** nodes.
- Rendered large graphs with GPU instancing after Unity's built-in batching broke down at scale, keeping the tool responsive.
- Applied Observer, Strategy and Factory patterns to split UI, input and layout code into modules that change on their own.
- Extended the interaction history to store each action's type and parameters, giving users undo and redo across their edits.
- Turned customer meetings into GitHub backlog items with acceptance tests, shipping Linux and Windows builds with manuals.
- Built a modular tutorial system for VR and desktop with in-app workload surveys and JSON telemetry of per-step task times.
- Ran an IRB-approved, counterbalanced **14**-person study comparing VR and desktop with SUS, NASA-TLX and think-aloud.

Graduate Teaching Assistant – CS 362 Software Engineering | Oregon State University Apr 2025 – Jun 2025
JavaScript, Vitest, Playwright, GitHub Actions *Corvallis, OR*

- Built Vitest and Playwright suites in GitHub Actions for **150+** students, cutting grading time by **70%** and saving **10+ hrs** a week.

Systems Engineer | Tata Consultancy Services Nov 2020 – Aug 2023
Python, SQL, Azure Data Factory, SQL Server, REST APIs, Azure DevOps, QlikSense *Hyderabad, India*

- Built Python and SQL pipelines in Azure Data Factory with full and incremental loads, moving **6+** on-premise sources into Azure.
- Replaced manual data transfers between dashboards and backend stores with REST APIs, cutting sync errors by **60%**.
- Added SIT, UAT and production validation gates to Azure DevOps CI/CD, cutting rollback incidents by **40%**.
- Set up row-level and sheet-level access controls so each of **500+** stakeholders saw only the data they were cleared for.
- Traced slow dashboard refreshes to queries and data models, then tuned indexes to cut refresh time from **20 min to under 5**.
- Built KPI dashboards with scheduled reports that replaced manual reporting, saving **$50K** a year.
- Handled post-production support, triaging issues reported by users and fixing data and dashboard defects.

Projects

NaVision: On-Device Computer Vision Assistant for Low-Vision Users | *C#, Unity, YOLOv8, ONNX, Unity Sentis, GPT-4o, Wit.ai*

- Ran YOLOv8 as ONNX on-device through Unity Sentis on Quest 3 camera frames, keeping detection latency under **150 ms**.
- Sent camera frames to GPT-4o for spoken scene descriptions, and added Wit.ai voice commands for hands-free use.
- Split vision, language and audio into providers behind shared interfaces, so each backend could be swapped on its own.
- Combined camera frames with 6DoF head tracking and camera intrinsics to place each detection at its true 3D position.

EchoRep: AR Shopping Assistant (team project, XRCC hackathon, showcased at IFA) | *C#, Unity, ZXing.Net*

- Built the QR code scanner on the Quest 3 passthrough camera with ZXing.Net, decoding product codes for the team's AI pipeline.
- Helped shape the product story and parts of the user flow with teammates working across several countries.
- Tested the final features end to end before the team presented at IFA Berlin, and later demoed it at OhioXR in Columbus.

Skills

Languages: C# (Unity/.NET), Python, SQL, JavaScript, TypeScript, C++
Systems & Deployment: Linux, command line, Docker, WSL2, Git, GitHub, REST APIs, Azure, Azure DevOps, CI/CD
Testing: Unit and integration testing, Playwright, Vitest, Unity Test Framework, regression testing, SIT/UAT, test case design
Vision, AI & Research: YOLOv8, ONNX, Unity Sentis, OpenAI API, pandas, scikit-learn, IRB-approved user studies, SUS, NASA-TLX